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Sacituzumab Tirumotecan in EGFR-TKI–Resistant, *EGFR*-Mutated Advanced NSCLC

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ABSTRACT

BACKGROUND

Sacituzumab tirumotecan (sac-TMT) is an antibody–drug conjugate targeting trophoblast cell-surface antigen 2 that has shown significant survival benefits in patients with *EGFR*-mutated non–small-cell lung cancer (NSCLC) that has progressed after epidermal growth factor receptor (EGFR) tyrosine kinase inhibitor (TKI) therapy and platinum-based chemotherapy.

METHODS

In this phase 3 trial, we enrolled patients with *EGFR*-mutated locally advanced or metastatic nonsquamous NSCLC that had progressed after EGFR-TKI therapy. The patients were randomly assigned, in a 1:1 ratio, to receive sac-TMT monotherapy or pemetrexed plus platinum-based chemotherapy. The primary end point was progression-free survival as assessed by blinded independent review. Overall survival was a hierarchically tested key secondary end point. In the interim analysis of progression-free survival as assessed by blinded independent review, sac-TMT monotherapy met the prespecified criterion for significance (two-sided P<0.0001); we report here the prespecified final analysis of progression-free survival and the preplanned interim analysis of overall survival.

RESULTS

Overall, 376 patients underwent randomization, with 188 assigned to each group. After a median follow-up of 18.9 months, the median progression-free survival was 8.3 months in the sac-TMT group and 4.3 months in the chemotherapy group (hazard ratio for disease progression or death, 0.49; 95% confidence interval [CI], 0.39 to 0.62). Overall survival was significantly longer with sac-TMT than with chemotherapy (hazard ratio for death, 0.60; 95% CI, 0.44 to 0.82; two-sided P=0.001); 18-month overall survival was 65.8% and 48.0%, respectively. Treatment-related adverse events of grade 3 or higher occurred in 58.0% of patients receiving sac-TMT and in 53.8% of those receiving chemotherapy, with the most common being a decreased neutrophil count (39.9% vs. 33.0%); treatment-related serious adverse events occurred in 9.0% and 17.6%, respectively.

CONCLUSIONS

In patients with *EGFR*-mutated advanced or metastatic NSCLC that had progressed after previous EGFR-TKI therapy, progression-free survival and overall survival outcomes were significantly better with sac-TMT than with platinum-based chemotherapy. (Funded by Sichuan Kelun-Biotech Biopharmaceutical; OptiTROP-Lung04 ClinicalTrials.gov number, NCT05870319.)

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A list of the OptiTROP-Lung04 trial Investigators is provided in the Supplementary Appendix, available at NEJM.org.

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CME

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Epidermal growth factor receptor (*EGFR*) mutations are present in a substantial proportion of patients with non–small-cell lung cancer (NSCLC).[1] For patients with *EGFR*-mutated advanced NSCLC who are not candidates for curative treatment or for those with metastatic disease, third-generation EGFR tyrosine kinase inhibitors (TKIs) are the standard first-line therapy.[2] However, acquired resistance to EGFR-TKIs inevitably develops, and subsequent treatment options are limited.[3,4] Platinum-based doublet chemotherapy remains a standard of care in this context, but its efficacy is modest.[5-7] Recent trials have explored chemotherapy-based combination strategies incorporating ivonescimab (an anti–programmed cell death protein 1 [PD-1] and vascular endothelial growth factor bispecific antibody),[8] sintilimab (a PD-1 inhibitor) with a bevacizumab biosimilar drug,[6] amivantamab (an EGFR and c-Met bispecific antibody) with or without lazertinib (an EGFR inhibitor for patients with *EGFR* exon 19 deletion),[9] and human epidermal growth factor receptor (HER) 3–directed antibody–drug conjugate.[10] These regimens have shown improvements in progression-free survival outcomes; however, overall survival benefits remain uncertain. Thus, there remains a need for new therapies for patients with EGFR-TKI–resistant, *EGFR*-mutated NSCLC.

Sacituzumab tirumotecan (sac-TMT) is an antibody–drug conjugate targeting trophoblast cell-surface antigen 2 (Trop-2) that was developed by conjugating the antibody to a belotecan-derived topoisomerase I inhibitor. Trop-2 is highly expressed in *EGFR*-mutated NSCLC and is associated with resistance to EGFR-TKIs.[11] Among various antigens that can be targeted with antibody–drug conjugates, Trop-2 is the most strongly and selectively expressed target in EGFR-TKI–resistant NSCLC cells.[12] Of note, the presence of *EGFR* mutations has been shown to lead to a significant increase in the internalization and lysosomal uptake of sac-TMT in NSCLC cells.[13] In the phase 1–2 KL264-01 study, sac-TMT showed encouraging antitumor activity in patients with EGFR-TKI–refractory advanced NSCLC who had not received chemotherapy.[13] These findings were further supported by the phase 2, randomized OptiTROP-Lung03 trial, in which sac-TMT showed a significant overall survival benefit over docetaxel in patients with *EGFR*-mutated NSCLC and progression after EGFR-TKI therapy and platinum-based chemotherapy.[14] This phase 3, confirmatory, randomized trial (OptiTROP-Lung04) was conducted to evaluate the efficacy and safety of sac-TMT in patients with *EGFR*-mutated and locally advanced or metastatic NSCLC that had progressed after EGFR-TKI therapy.

METHODS

TRIAL DESIGN AND PATIENTS

We conducted a phase 3, multicenter, open-label, randomized trial at 66 sites in China. Patients 18 to 75 years of age with histologically or cytologically confirmed locally advanced (stage IIIB or IIIC) or metastatic (stage IV) nonsquamous NSCLC who were not candidates for curative surgery or definitive chemoradiotherapy were eligible to enroll in the trial. All the patients had tumors harboring sensitizing *EGFR* mutations, including exon 19 deletions or exon 21 codon p.Leu858Arg (L858R) substitutions. Eligibility criteria required either progression after first- or second-generation EGFR-TKIs with documented T790M-negative status or progression after third-generation EGFR-TKIs regardless of T790M status. A score of 0 or 1 on the Eastern Cooperative Oncology Group (ECOG) performance-status scale (range, 0 to 5, with higher scores indicating greater disability) was also required. Patients with brain metastases were eligible if the lesions had been treated and were stable. Full eligibility criteria are specified in the protocol, available with the full text of this article at NEJM.org.

PROCEDURES

The patients were randomly assigned with the use of a centralized stratified block randomization system in a 1:1 ratio to receive either sac-TMT or pemetrexed plus platinum-based chemotherapy. Stratification factors included previous treatment with third-generation EGFR-TKIs (first-line use, second-line use, or no previous use) and brain metastases (presence or absence). Sac-TMT was administered intravenously at a dose of 5 mg per kilogram of body weight on day 1 and day 15 of each 28-day cycle. The chemotherapy regimen consisted of pemetrexed

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(500 mg per square meter of body-surface area) plus the investigator's choice of carboplatin (area under the curve of 5 mg per milliliter per minute) or cisplatin (75 mg per square meter), all administered on day 1 of each 21-day cycle for up to four cycles, followed by pemetrexed maintenance therapy. Treatment was continued until the occurrence of disease progression, unacceptable toxic effects, patient withdrawal, or the fulfillment of any other protocol-specified discontinuation criteria. Crossover between trial groups was not allowed; however, patients who were assigned to receive chemotherapy could receive sac-TMT after disease progression, since sac-TMT has been approved and is commercially available. Interventions and dose reductions to alleviate symptoms were permitted (details are provided in Supplementary Methods in the Supplementary Appendix, available at NEJM.org).

ASSESSMENTS

Tumor response and progression were evaluated by blinded independent review according to Response Evaluation Criteria in Solid Tumors (RECIST), version 1.1. Adverse events were monitored throughout the trial, coded according to the *Medical Dictionary for Regulatory Activities*, version 26.0, and graded according to the National Cancer Institute Common Terminology Criteria for Adverse Events, version 5.0. Adverse events that emerged during treatment were defined as all adverse medical events that occurred or worsened from the time the patient first received the trial treatment until 30 days after the last dose or before the patient started new antitumor therapy (whichever occurred first). Patient-reported outcomes were evaluated with the validated European Organization for Research and Treatment of Cancer (EORTC) quality-of-life core questionnaire along with its lung cancer module.[15-17]

END POINTS

The primary end point was progression-free survival as assessed by blinded independent review according to RECIST, version 1.1. Secondary end points included overall survival; progression-free survival as assessed by investigators; objective response (complete or partial response), disease control, and response duration (each assessed by both blinded independent review and investigators); safety; and patient-reported outcomes.

TRIAL OVERSIGHT

This trial was designed by the sponsor, Sichuan Kelun-Biotech Biopharmaceutical, in collaboration with the authors. The protocol and all the amendments were approved by the institutional review board of each participating site. Written informed consent was obtained from all the participants before enrollment. The trial was conducted in accordance with the principles of the Declaration of Helsinki and all applicable local regulations. An independent data monitoring committee oversaw trial conduct, safety, and outcomes. The authors and the sponsor vouch for the completeness and accuracy of the data and for the fidelity of the trial to the protocol. The first draft of the manuscript was prepared by authors who are not employees of the sponsor in collaboration with the sponsor, with medical writing support funded by the sponsor. Both the sponsor and the authors were involved in the collection, analysis, and interpretation of the data, and they reviewed and approved the manuscript that was submitted for publication.

STATISTICAL ANALYSIS

We calculated, on the basis of historical data, that approximately 356 participants would have to be enrolled for a total of 249 events (disease progression or death) to occur to give the trial 88% power to detect a hazard ratio of 0.67 for progression-free survival at a one-sided alpha level of 0.025.[6,18] This sample size would also provide the trial with 80% power to detect the difference between the groups in overall survival with the assumption of a hazard ratio of 0.70. Details on the sample size calculation can be found in the protocol.

An interim analysis of progression-free survival was planned to be performed after at least 174 events had been observed. A Lan–DeMets alpha-spending function was used to estimate the O'Brien–Fleming boundary to control the overall type I error. Overall survival was hierarchically tested, and an interim analysis with the use of a similar alpha-spending approach was planned to be performed after 188 deaths had occurred or 24 months after the first participant underwent randomization.

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The interim analysis of progression-free survival was conducted with the data-cutoff date of July 11, 2024; the actual alpha level at the interim analysis was 0.0337 (two-sided). The results of the interim analysis were reviewed by the independent data monitoring committee, which reported that the significance boundary for progression-free survival had been met (two-sided P<0.0001) (Fig. S1). As of the data-cutoff date of July 6, 2025, the preplanned interim analysis of overall survival (actual boundary, two-sided 0.0124, on the basis of 168 deaths observed) and the final analysis of progression-free survival were conducted, the results of which are shown here.

Efficacy analyses were conducted in the intention-to-treat population, which included all the patients who underwent randomization. The safety analysis included all the patients who received at least one dose of trial treatment and had safety assessment data. The Kaplan–Meier method was used to estimate median survival. Stratified Cox proportional-hazards models were used to calculate hazard ratios and corresponding 95% confidence intervals. Between-group comparisons of survival were performed with stratified log-rank tests. Stratified analyses were based on the factors used at randomization. For time-to-event end points, missing data were handled under the assumption of noninformative censoring (see Table S1 in the Supplementary Appendix for detailed censoring rules).

The percentage of patients with an objective response was calculated with exact (Clopper–Pearson) confidence intervals, and between-group differences were assessed with the Cochran–Mantel–Haenszel test with adjustment for stratification factors. In the case of missing data for tumor response in patients who did not receive treatment, who had no postbaseline assessments, or who had imaging that could not be evaluated, the patients were conservatively assumed not to have had a response. Missing data for other end points were assumed to be missing at random. Confidence intervals for end points that were not part of the hypothesis testing were not adjusted for multiplicity and should be interpreted as descriptive only. Prespecified subgroup analyses and a supplementary analysis are detailed in the statistical analysis plan (Supplementary Appendix). All statistical analyses were performed with the use of SAS software, version 9.4 (SAS Institute).

RESULTS

PATIENTS AND TREATMENT

Between July 6, 2023, and April 24, 2024, a total of 596 patients were screened, of whom 376 were enrolled and randomly assigned to receive sac-TMT (188 patients) or chemotherapy (188 patients) (Fig. S2). The median age of the patients was 60 years (range, 31 to 75), and 39.6% were men. The majority of the patients (74.5%) had never smoked. Most of the patients (97.6%) had stage IV disease at enrollment. All the patients had an ECOG performance-status score of 0 or 1 (20.7% and 79.3%, respectively). Most of the patients (94.7%) had received previous third-generation EGFR-TKIs, particularly as first-line therapy (in 62.5%). The baseline characteristics of the patients were generally balanced between the two groups (Table 1). This trial was conducted in China and enrolled only Asian persons. The characteristics of the patients were largely representative of the broad population of Chinese patients with *EGFR*-mutated NSCLC (Table S2).

As of the data-cutoff date of July 6, 2025, a total of 40 patients (21.3%) in the sac-TMT group and 3 patients (1.6%) in the chemotherapy group were still receiving trial treatments (Fig. S2). All the patients were included in the intention-to-treat population, whereas the safety population included 188 patients in the sac-TMT group and 182 patients in the chemotherapy group (6 patients were excluded because they underwent randomization but did not receive trial treatment).

PROGRESSION-FREE SURVIVAL

After a median follow-up of 18.9 months, disease progression (as assessed by blinded independent review) or death had occurred in 144 patients (76.6%) in the sac-TMT group and in 159 patients (84.6%) in the chemotherapy group. The median progression-free survival as assessed by blinded independent review was longer among the patients in the sac-TMT group (8.3 months; 95% confidence interval [CI], 6.7 to 9.9) than among those in the chemotherapy group (4.3 months; 95% CI, 4.2 to 5.5), with a hazard ratio for disease progression or death of 0.49 (95% CI,

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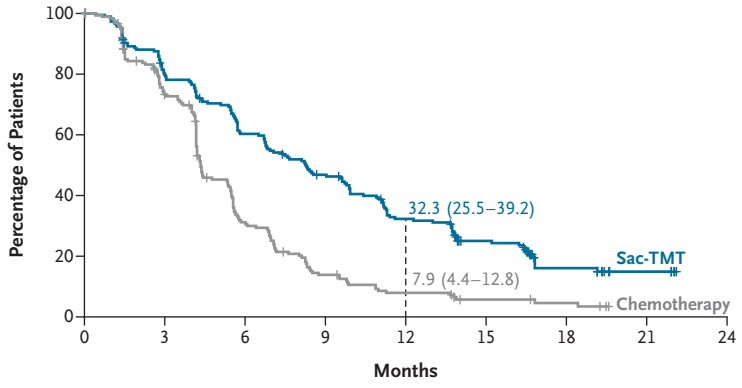
Table 1. Baseline Characteristics of the Patients (intention-to-Treat Population).*

Characteristic	Sacituzumab Tirumotecan (N = 188)	Chemotherapy (N = 188)
Age		
Median (range) — yr	60 (31–75)	59 (33–75)
≥65 yr — no. (%)	58 (30.9)	51 (27.1)
Male sex — no. (%)	66 (35.1)	83 (44.1)
Asian race — no. (%)	188 (100.0)	188 (100.0)
Smoking history — no. (%)		
Current or former smoker	43 (22.9)	53 (28.2)
Never smoked	145 (77.1)	135 (71.8)
ECOG performance-status score — no. (%)†		
0	35 (18.6)	43 (22.9)
1	153 (81.4)	145 (77.1)
Adenocarcinoma — no. (%)	188 (100.0)	188 (100.0)
Disease stage — no. (%)‡		
IIIB or IIIC	6 (3.2)	3 (1.6)
IV	182 (96.8)	185 (98.4)
Brain metastases — no. (%)	33 (17.6)	36 (19.1)
Liver metastases — no. (%)	25 (13.3)	33 (17.6)
≥3 Metastatic sites — no. (%)	128 (68.1)	126 (67.0)
EGFR mutation subtype — no. (%)§		
Exon 21 L858R substitution	84 (44.7)	71 (37.8)
Exon 19 deletion	106 (56.4)	118 (62.8)
Other	8 (4.3)	7 (3.7)
T790M mutation status — no. (%)¶		
Negative	48 (25.5)	40 (21.3)
Positive	29 (15.4)	36 (19.1)
Unknown	111 (59.0)	112 (59.6)
Previous third-generation EGFR-TKI — no.(%)‖	178 (94.7)	178 (94.7)
First-line therapy	118 (62.8)	117 (62.2)
Second-line therapy	60 (31.9)	60 (31.9)

* The intention-to-treat population included all the patients who underwent randomization. TKI denotes tyrosine kinase inhibitor.
† Eastern Cooperative Oncology Group (ECOG) performance-status scores range from 0 to 5, with higher scores indicating greater disability.
‡ Disease stage was determined according to the American Joint Committee on Cancer, 8th edition, tumor–node–metastasis staging system.
§ The sum of percentages exceeds 100 because of overlapping mutations in some patients. The subcategory "Other" includes other *EGFR* mutations co-occurring with exon 21 L858R substitution or exon 19 deletion.
¶ The high percentage of patients with unknown T790M status reflects the eligibility criterion allowing patients to be enrolled if their disease had progressed after they had received treatment with third-generation EGFR-TKIs regardless of T790M mutation status. Many such patients lacked T790M testing results.
‖ The total number of patients who received previous treatment with third-generation EGFR-TKIs (356) does not equal the sum of patients who received first-line or second-line third-generation EGFR-TKIs (355) because one patient in the chemotherapy group had previously received osimertinib but discontinued it owing to toxic effects. This patient's disease later progressed after treatment with a second-generation EGFR-TKI, and the patient was enrolled in the trial (T790M mutation negative). Consequently, this patient is categorized as having previous third-generation EGFR-TKI use but is not included in the count for first-line or second-line third-generation EGFR-TKI therapy.

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A Progression-free Survival



	No. of Events/ No. of Patients (%)	Median Progression-free Survival (95% CI)
		mo
Sac-TMT	144/188 (76.6)	8.3 (6.7–9.9)
Chemotherapy	159/188 (84.6)	4.3 (4.2–5.5)

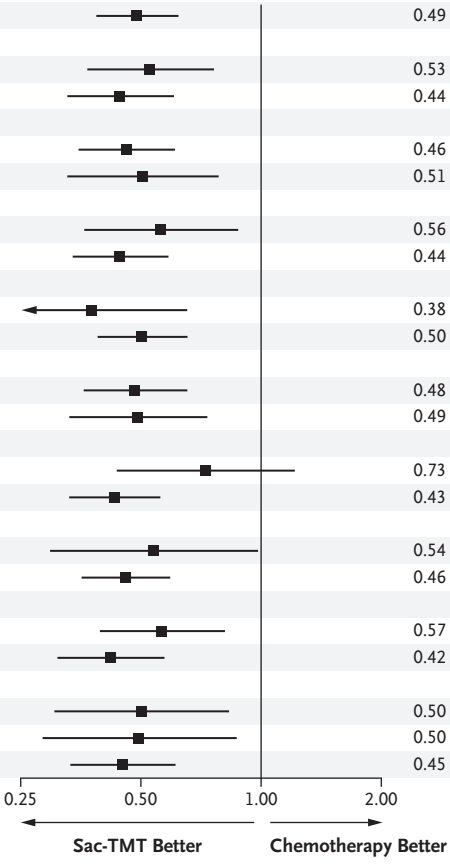
Hazard ratio for disease progression or death, 0.49 (95% CI, 0.39–0.62)

No. at Risk

Sac-TMT	188	144	108	82	55	35	14	5	0
Chemotherapy	188	125	51	22	12	6	4	0	

B Analysis of Progression-free Survival in Prespecified Subgroups

Subgroup	Sac-TMT	Chemotherapy	Hazard Ratio for Disease Progression or Death (95% CI)
	no. of events/total no. of patients		
All patients	144/188	159/188	0.49 (0.39–0.62)
Sex			
Male	54/66	75/83	0.53 (0.37–0.76)
Female	90/122	84/105	0.44 (0.33–0.60)
Age			
<65 yr	100/130	118/137	0.46 (0.35–0.61)
≥65 yr	44/58	41/51	0.51 (0.33–0.78)
History of smoking			
Current or former smoker	36/43	49/53	0.56 (0.36–0.88)
Never smoked	108/145	110/135	0.44 (0.34–0.59)
ECOG performance-status score			
0	24/35	35/43	0.38 (0.22–0.65)
1	120/153	124/145	0.50 (0.39–0.65)
Previous third-generation EGFR-TKI therapy			
First-line therapy	89/118	95/117	0.48 (0.36–0.65)
Second-line therapy	49/60	55/60	0.49 (0.33–0.73)
Brain metastases			
Yes	30/33	31/36	0.73 (0.43–1.21)
No	114/155	128/152	0.43 (0.33–0.56)
Liver metastases			
Yes	18/25	28/33	0.54 (0.30–0.98)
No	126/163	131/155	0.46 (0.35–0.59)
EGFR mutation subtype			
Exon 21 L858R substitution	68/84	59/71	0.57 (0.39–0.81)
Exon 19 deletion	77/106	101/118	0.42 (0.31–0.57)
T790M mutation status			
Negative	39/48	32/40	0.50 (0.30–0.83)
Positive	23/29	30/36	0.50 (0.28–0.87)
Unknown	82/111	97/112	0.45 (0.33–0.61)



0.39 to 0.62). Progression-free survival at 12 months was 32.3% (95% CI, 25.5 to 39.2) in the sac-TMT group and 7.9% (95% CI, 4.4 to 12.8) in the chemotherapy group (Fig. 1A). The most common sites of progression were lung and central nervous system (Table S3). Results according to investigator assessment were consistent with those by blinded independent review, with a me-

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Figure 1 (facing page). Final Analysis of Progression-free Survival.

Shown are the results in the intention-to-treat population (all 376 patients who underwent randomization) as of the data-cutoff date of July 6, 2025. Panel A shows Kaplan–Meier estimates for the final analysis of progression-free survival as assessed by blinded independent review. Panel B shows a forest plot of progression-free survival as assessed by blinded independent review in prespecified subgroups.

Progression-free survival was defined as the time from randomization to the first documented disease progression or death from any cause. Tumor assessments were conducted by blinded independent review according to Response Evaluation Criteria in Solid Tumors, version 1.1. Eastern Cooperative Oncology Group (ECOG) performance-status scores range from 0 to 5, with higher scores indicating greater disability. Confidence intervals have not been adjusted for multiplicity and should not be used in place of hypothesis testing. Sac-TMT denotes sacituzumab tirumotecan.

dian progression-free survival of 8.4 months (95% CI, 7.1 to 9.7) among the patients in the sac-TMT group and 4.8 months (95% CI, 4.2 to 5.5) among those in the chemotherapy group, and with a hazard ratio of 0.51 (95% CI, 0.41 to 0.65). Progression-free survival at 12 months according to investigator assessment was 34.7% (95% CI, 27.7 to 41.7) and 10.7% (95% CI, 6.5 to 16.0), respectively (Fig. S3A). The progression-free survival benefit with sac-TMT appeared to be consistent across most of the prespecified subgroups; however, the confidence intervals crossing 1.00 should be interpreted with caution because of the limited size of the subgroups (Fig. 1B and Fig. S3B).

OVERALL SURVIVAL

At the preplanned interim analysis of overall survival, 67 deaths (35.6% maturity) had occurred in the sac-TMT group and 101 deaths (53.7% maturity) had occurred in the chemotherapy group after a median follow-up of 18.9 months. Overall survival was significantly longer with sac-TMT than with chemotherapy (median, could not be estimated vs. 17.4 months), with a hazard ratio for death of 0.60 (95% CI, 0.44 to 0.82) and a two-sided P value of 0.001, which crossed the prespecified significance boundary (two-sided alpha of 0.0124). Overall survival at 18 months was 65.8% (95% CI, 58.3 to 72.3) with sac-TMT and 48.0% (95% CI, 40.2 to 55.4) with chemotherapy (Fig. 2A).

Of the patients who discontinued the trial treatment (148 patients in the sac-TMT group and 179 in the chemotherapy group), 72.3% and 85.5%, respectively, received subsequent anticancer treatment. Among the patients who discontinued the trial treatment, 41.9% of those in the sac-TMT group and 53.6% of those in the chemotherapy group received subsequent chemotherapy; 37.2% and 12.8%, respectively, received pemetrexed-based chemotherapy. Subsequent antibody–drug conjugates were used in 1.4% of the patients in the sac-TMT group and in 19.6% of those in the chemotherapy group (Table S4). In the supplementary analysis, in which data were censored at the start date of subsequent antibody–drug conjugate treatment, the hazard ratio for death was 0.56 (95% CI, 0.41 to 0.77), and overall survival at 18 months was 66.1% (95% CI, 58.6 to 72.6) in the sac-TMT group and 44.6% (95% CI, 35.9 to 52.9) in the chemotherapy group (Fig. S4).

Analyses in the prespecified subgroups are shown in Figure 2B. Among the patients with brain metastases, the hazard ratio for death was 0.65 (95% CI, 0.32 to 1.30); among those with liver metastases, the hazard ratio was 0.53 (95% CI, 0.26 to 1.10). The wide confidence intervals for both of these subgroups (brain and liver metastases) were a reflection, in part, of the small numbers of patients.

RESPONSE

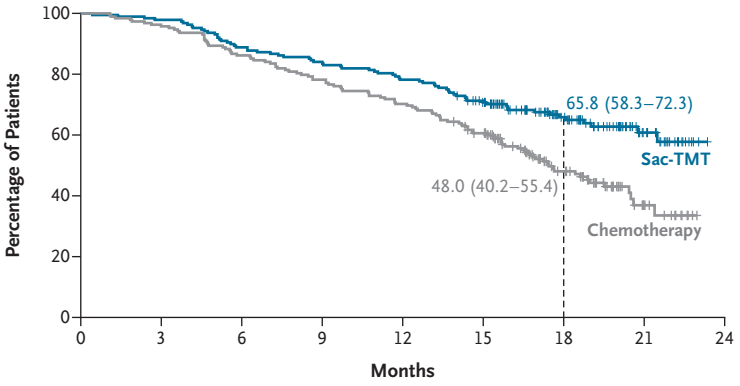
The percentage of patients with an objective response as assessed by blinded independent review was 60.6% (95% CI, 53.3 to 67.7) in the sac-TMT group and 43.1% (95% CI, 35.9 to 50.5) in the chemotherapy group, which yielded an estimated between-group difference of 17.0 percentage points (95% CI, 7.0 to 27.1) (Table 2). The median duration of response was 8.3 months (95% CI, 6.2 to 10.0) with sac-TMT and 4.2 months (95% CI, 3.0 to 4.4) with chemotherapy, and the estimated percentage of patients with a response duration of at least 12 months was 36.3% (95% CI, 27.3 to 45.3) and 8.1% (95% CI, 3.3 to 15.8), respectively (Fig. S5). These findings were consistent with those according to investigator assessment (Table S5).

SAFETY

In the safety population, as of the data-cutoff date of July 6, 2025, the median duration of treatment was 9.6 months (range, 0.5 to 23.5) in

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A Overall Survival



	No. of Events/ No. of Patients (%)	Median Overall Survival (95% CI)
		mo
Sac-TMT	67/188 (35.6)	NE (21.5–NE)
Chemotherapy	101/188 (53.7)	17.4 (15.7–20.4)

Hazard ratio for death, 0.60 (95% CI, 0.44–0.82)
Two-sided P=0.001 by stratified log-rank test
Boundary for statistical significance, two-sided alpha, 0.0124

No. at Risk

Sac-TMT	188	184	167	158	147	127	75	25	0
Chemotherapy	188	180	162	147	132	110	57	13	0

B Analysis of Overall Survival in Prespecified Subgroups

Subgroup	Sac-TMT	Chemotherapy	Hazard Ratio for Death (95% CI)
	no. of events/total no. of patients		
All patients	67/188	101/188	0.60 (0.44–0.82)
Sex			
Male	28/66	47/83	0.70 (0.44–1.11)
Female	39/122	54/105	0.54 (0.36–0.81)
Age			
<65 yr	41/130	69/137	0.55 (0.37–0.81)
≥65 yr	26/58	32/51	0.62 (0.37–1.05)
History of smoking			
Current or former smoker	18/43	29/53	0.71 (0.40–1.29)
Never smoked	49/145	72/135	0.55 (0.38–0.79)
ECOG performance-status score			
0	10/35	19/43	0.57 (0.27–1.23)
1	57/153	82/145	0.59 (0.42–0.83)
Previous third-generation EGFR-TKI therapy			
First-line therapy	44/118	67/117	0.59 (0.40–0.86)
Second-line therapy	20/60	29/60	0.58 (0.33–1.03)
Brain metastases			
Yes	13/33	21/36	0.65 (0.32–1.30)
No	54/155	80/152	0.57 (0.40–0.81)
Liver metastases			
Yes	12/25	22/33	0.53 (0.26–1.10)
No	55/163	79/155	0.60 (0.43–0.85)
EGFR mutation subtype			
Exon 21 L858R substitution	36/84	38/71	0.75 (0.48–1.18)
Exon 19 deletion	31/106	63/118	0.46 (0.30–0.71)
T790M mutation status			
Negative	18/48	21/40	0.67 (0.35–1.25)
Positive	7/29	20/36	0.36 (0.15–0.85)
Unknown	42/111	60/112	0.62 (0.42–0.92)

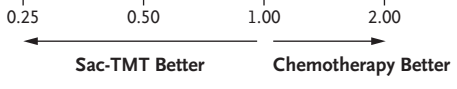


Figure 2. Interim Analysis of Overall Survival.

Shown are the results from the intention-to-treat population (all 376 patients who underwent randomization) as of the data-cutoff date of July 6, 2025. Panel A shows Kaplan–Meier estimates of overall survival. Panel B shows a forest plot of overall survival in prespecified subgroups. Overall survival was defined as the time from randomization to death from any cause. For Panel B, the confidence intervals have not been adjusted for multiplicity and should not be used in place of hypothesis testing. NE denotes could not be estimated.

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Table 2. Efficacy End Points as Assessed by Blinded Independent Review (Intention-to-Treat Population).*

End Point	Sacituzumab Tirumotecan (N=188)	Chemotherapy (N=188)	Difference (95% CI)
Best overall response — no. (%)			
Complete response	1 (0.5)	0	
Partial response	113 (60.1)	81 (43.1)	
Stable disease	50 (26.6)	70 (37.2)	
Progressive disease	20 (10.6)	26 (13.8)	
Imaging could not be evaluated	1 (0.5)	0	
No assessment†	3 (1.6)	11 (5.9)	
Objective response — no. (% [95% CI])‡	114 (60.6 [53.3–67.7])	81 (43.1 [35.9–50.5])	17.0 (7.0 to 27.1)
Disease control — no. (% [95% CI])§	164 (87.2 [81.6–91.6])	151 (80.3 [73.9–85.7])	6.7 (–0.7 to 14.0)
Median response duration (95% CI) — mo¶	8.3 (6.2–10.0)	4.2 (3.0–4.4)	
Response duration ≥12 mo (95% CI) — %‖	36.3 (27.3–45.3)	8.1 (3.3–15.8)	

* Tumor response was evaluated by blinded independent review according to the Response Evaluation Criteria in Solid Tumors, version 1.1. The data-cutoff date was July 6, 2025. The intention-to-treat population included all the patients who underwent randomization. Confidence intervals have not been adjusted for multiplicity and should not be used in place of hypothesis testing. The between-group difference was estimated with the Cochran–Mantel–Haenszel test.
† Three patients in the sac-TMT group and 11 in the chemotherapy group had no postbaseline tumor imaging assessments. Among them, 6 patients in the chemotherapy group did not receive treatment and therefore had no post-treatment tumor evaluations.
‡ Objective response was defined as a complete or partial response.
§ Disease control was defined as a complete response, partial response, or stable disease.
¶ Response duration was defined as the time from the first documentation of response to disease progression or death, whichever occurred first. The median response duration was estimated with the Kaplan–Meier method, with its 95% confidence interval calculated by the Brookmeyer–Crowley method.
‖ The percentages of patients with a response duration of at least 12 months were estimated with the Kaplan–Meier method, and the corresponding 95% confidence intervals were estimated with the exponential Greenwood formula.

the sac-TMT group and 4.9 months (range, 0.7 to 22.6) in the chemotherapy group (Table S6). Adverse events that emerged during treatment are shown in Table S7. Most of the patients had treatment-related adverse events (100% in the sac-TMT group and 98.4% in the chemotherapy group). Treatment-related adverse events of grade 3 or higher were reported in 58.0% of the patients in the sac-TMT group and in 53.8% of those in the chemotherapy group (Table 3). Treatment-related serious adverse events occurred less frequently in the sac-TMT group than in the chemotherapy group (9.0% vs. 17.6%); details of the events are shown in Table S8. The treatment-related adverse event of grade 3 or higher that occurred most frequently in both groups was a decreased neutrophil count (39.9% with sac-TMT vs. 33.0% with chemotherapy). The use of granulocyte colony-stimulating factor was recorded in 59.0% of the patients in the sac-TMT group and in 57.7% of those in the chemotherapy group. Febrile neutropenia occurred in one patient (0.5%) in the sac-TMT group and in five patients (2.7%) in the chemotherapy group. Sac-TMT was associated with a lower incidence of anemia of grade 3 or higher than chemotherapy (11.2% vs. 14.3%), as well as a lower incidence of thrombocytopenia of grade 3 or higher (2.1% vs. 16.5%). None of the treatment-related adverse events led to treatment discontinuation in the sac-TMT group (Table 3).

Regarding drug-related adverse events of special interest (Table S9), the sac-TMT group had a higher incidence of stomatitis than the chemotherapy group (64.4% vs. 4.9%). In the sac-TMT group, grade 1 stomatitis occurred in 42 patients (22.3%), grade 2 in 70 patients (37.2%), and grade 3 in 9 patients (4.8%); no cases of grade 4

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Table 3. Treatment-Related Adverse Events (Safety Population).*

Event	Sacituzumab Tirumotecan (N=188)		Chemotherapy (N=182)	
	Any grade	Grade ≥3	Any grade	Grade ≥3
	number of patients (percent)			
Any treatment-related adverse event	188 (100.0)	109 (58.0)	179 (98.4)	98 (53.8)
Leading to dose reduction	57 (30.3)	—	41 (22.5)	—
Leading to dose interruption	69 (36.7)	—	60 (33.0)	—
Leading to treatment discontinuation	0	—	1 (0.5)	—
Leading to death†	0	—	1 (0.5)	—
Any treatment-related serious adverse event	17 (9.0)	—	32 (17.6)	—
Treatment-related adverse event with an incidence of ≥10% in either group				
Anemia	159 (84.6)	21 (11.2)	139 (76.4)	26 (14.3)
White-cell decreased	157 (83.5)	52 (27.7)	127 (69.8)	40 (22.0)
Alopecia	157 (83.5)	0	17 (9.3)	0
Neutrophil count decreased	142 (75.5)	75 (39.9)	126 (69.2)	60 (33.0)
Stomatitis‡	121 (64.4)	9 (4.8)	9 (4.9)	0
Nausea	89 (47.3)	1 (0.5)	86 (47.3)	2 (1.1)
Anorexia	78 (41.5)	0	58 (31.9)	0
Fatigue	72 (38.3)	7 (3.7)	73 (40.1)	4 (2.2)
Weight loss	52 (27.7)	0	28 (15.4)	1 (0.5)
Thrombocytopenia	51 (27.1)	4 (2.1)	85 (46.7)	30 (16.5)
Vomiting	50 (26.6)	0	39 (21.4)	1 (0.5)
Alanine aminotransferase increased	46 (24.5)	1 (0.5)	63 (34.6)	2 (1.1)
Constipation	39 (20.7)	0	31 (17.0)	0
Aspartate aminotransferase increased	35 (18.6)	1 (0.5)	63 (34.6)	2 (1.1)
Rash	35 (18.6)	0	14 (7.7)	0
Lymphocyte count decreased	30 (16.0)	6 (3.2)	23 (12.6)	7 (3.8)
Hypoalbuminemia	23 (12.2)	0	27 (14.8)	0
γ-Glutamyltransferase increased	20 (10.6)	2 (1.1)	27 (14.8)	3 (1.6)
Hyperuricemia	20 (10.6)	0	17 (9.3)	0
Diarrhea	19 (10.1)	1 (0.5)	6 (3.3)	0
Hypokalemia	14 (7.4)	4 (2.1)	23 (12.6)	7 (3.8)

* The data-cutoff date for the safety analysis was July 6, 2025. The safety population included all the patients who received at least one dose of the assigned trial treatment and underwent a safety assessment. Adverse events were coded according to the *Medical Dictionary for Regulatory Activities*, version 26.0, and graded with the use of the National Cancer Institute Common Terminology Criteria for Adverse Events, version 5.0. Adverse events that emerged during treatment were defined as all adverse medical events that occurred or worsened from the time the patient first received the trial treatment until 30 days after the last dose or before the patient started new antitumor therapy (whichever occurred first).

† One patient (0.5%) in the chemotherapy group died from a treatment-related adverse event (cardiorespiratory arrest).

‡ Stomatitis was reported as a group term that included stomatitis, oral ulcer, and aphthous ulcer. In the sac-TMT group, grade 1 stomatitis occurred in 42 patients (22.3%), grade 2 in 70 (37.2%), and grade 3 in 9 (4.8%); no cases of grade 4 or 5 were reported. In the chemotherapy group, 7 patients (3.8%) had grade 1 stomatitis and 2 (1.1%) had grade 2 stomatitis.

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or 5 were reported. A total of 19 patients (10.1%) in the sac-TMT group had dose reductions because of stomatitis, but no discontinuations occurred because of stomatitis. With appropriate interventions and dose reductions to alleviate symptoms, all nine cases of grade 3 stomatitis improved to grade 2 or lower without treatment discontinuation within a median of 10 days after diagnosis. Ocular-surface toxic effects (e.g., dry eye syndrome, increased lacrimation, and keratitis) occurred in 9.6% of the patients in the sac-TMT group and in 0.5% of those in the chemotherapy group, with no cases of grade 3 or higher (Table S9). Only one infusion-related reaction (grade 2) was reported, which was in the sac-TMT group. No drug-related interstitial lung disease or pneumonitis was reported in the sac-TMT group.

PATIENT-REPORTED OUTCOMES

Among the patients who underwent randomization, 97.3% of the patients in the sac-TMT group and 89.9% of those in the chemotherapy group were able to be evaluated for quality of life. The time to a deterioration of quality of life related to global health status and to a deterioration of clinical status defined by the presence of key lung cancer–related symptoms including dyspnea, cough, and chest pain over the duration of trial treatment appeared to be longer with sac-TMT than with chemotherapy (Table S10 and Fig. S6). The curves for these patient-reported outcomes showed a slight overlap during the initial 3-month period and consistent separation thereafter.

DISCUSSION

In this phase 3, randomized trial, sac-TMT led to significantly better progression-free survival and overall survival outcomes than platinum-based chemotherapy in patients with EGFR-TKI–resistant, *EGFR*-mutated advanced NSCLC. These benefits appeared to be generally consistent across the subgroups. Sac-TMT was associated with a lower incidence of treatment-related serious adverse events and hematologic toxic effects, such as anemia and thrombocytopenia, than chemotherapy. No new safety signals were identified.

Recent therapeutic advances have primarily involved multiagent regimens combining chemotherapy with an immune checkpoint inhibitor, antiangiogenic agent, or bispecific EGFR and c-Met antibody or a HER3-directed antibody–drug conjugate, as investigated in the HARMONi-A, ORIENT-31, MARIPOSA-2, and HERTHENA-Lung02 trials, respectively, but the overall survival benefit did not reach significance.[6,8-10] Of note, sac-TMT monotherapy in our trial showed a significant benefit at the preplanned interim analysis of overall survival after a median follow-up of 18.9 months.

Recent in vitro experiments showed that the presence of *EGFR* mutations in NSCLC cells markedly increased sac-TMT internalization and lysosomal uptake as compared with wild-type cells.[13] *EGFR*-mutated NSCLC cells with EGFR-TKI resistance showed even greater sac-TMT uptake than cells that had not received a TKI. This enhanced internalization further translated into higher antitumor activity in *EGFR*-mutated than in wild-type patient-derived lung cancer organoids.[13] These findings contribute to an understanding of the significant survival benefit observed with sac-TMT as compared with docetaxel in the OptiTROP-Lung03 trial, which specifically enrolled patients with NSCLC harboring sensitizing *EGFR* mutations,[14] whereas other Trop-2-antibody–drug conjugates have not shown a significant survival benefit in unselected NSCLC populations with or without actionable genomic alterations.[19,20] The overall survival benefit shown in the present trial further supports the potent antitumor efficacy of sac-TMT in *EGFR*-mutated NSCLC in the context of EGFR-TKI resistance.

Analyses in prespecified subgroups showed patterns favoring sac-TMT for both progression-free and overall survival, including among patients with brain metastases, a population with particularly limited treatment options and an unfavorable prognosis.[21] Since first- and second-generation EGFR-TKIs remain a recommendation for first-line therapy according to the Chinese guidelines, a small proportion of patients in our trial with negative T790M mutation status had received first- or second-generation EGFR-TKIs. Treatment efficacy with sac-TMT was seen in patients whose tumors had progressed after previous treatment with third-generation EGFR-TKIs — the current standard for *EGFR*-mutated advanced NSCLC — either as first-line or second-line therapy.[22-24] These collective findings rein-

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force sac-TMT as an effective therapeutic option for the broad population of patients with *EGFR*-mutated advanced NSCLC with progression after EGFR-TKIs.

The safety results are consistent with those in previous reports of sac-TMT and suggest that no new safety signals were observed.[13,14,25,26] Sac-TMT was associated with a higher incidence of stomatitis than chemotherapy, with most cases being mild and with grade 3 or higher cases reported in very few patients (4.8%; all were grade 3). Furthermore, the time to deterioration of global quality of life was longer with sac-TMT than with chemotherapy, which indicated that the adverse events did not compromise global quality of life. With sufficient premedication, only one infusion-related reaction (grade 2) was reported in the sac-TMT group. Other drug-related adverse events of special interest were rare, with a low incidence of ocular-surface toxic effects, and no interstitial lung disease or pneumonitis was observed.

A strength of our trial is that the results for overall survival reached statistical significance. A limitation is that although this trial used a multicenter design, it was conducted in one region and enrolled only Asian persons, which raises a question regarding the ability to extrapolate these outcomes to other patient groups. An exploration of biomarkers in *EGFR*-mutated NSCLC may be valuable for determining patient selection,[27] and future studies may focus on this question.

Currently, other active agents, such as amivantamab or ivonescimab, are typically administered in combination with chemotherapy and have shown limited or uncertain overall survival benefits. On the other hand, sac-TMT was administered as monotherapy in this trial and has shown significant progression-free survival and overall survival benefits over pemetrexed plus platinum-based chemotherapy. Therefore, sac-TMT could be considered to be a favorable treatment option before pemetrexed plus platinum-based chemotherapy for EGFR-TKI–resistant NSCLC is considered.

Among patients with advanced *EGFR*-mutated NSCLC that had progressed after EGFR-TKI therapy, progression-free survival and overall survival outcomes were significantly better with sac-TMT than with standard pemetrexed plus platinum-based chemotherapy.

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REFERENCES

1. Melosky B, Kambartel K, Häntschel M, et al. Worldwide prevalence of epidermal growth factor receptor mutations in non-small cell lung cancer: a meta-analysis. Mol Diagn Ther 2022;26:7-18.
2. Hendriks LEL, Remon J, Faivre-Finn C, et al. Non-small-cell lung cancer. Nat Rev Dis Primers 2024;10:71.
3. Zhou F, Guo H, Xia Y, et al. The changing treatment landscape of EGFR-mutant non-small-cell lung cancer. Nat Rev Clin Oncol 2025;22:95-116.
4. Fu K, Xie F, Wang F, Fu L. Therapeutic strategies for EGFR-mutated non-small cell lung cancer patients with osimertinib resistance. J Hematol Oncol 2022;15:173.
5. Mok TSK, Kim S-W, Wu Y-L, et al. Gefitinib plus chemotherapy versus chemotherapy in epidermal growth factor receptor mutation-positive non-small-cell lung cancer resistant to first-line gefitinib (IMPRESS): overall survival and biomarker analyses. J Clin Oncol 2017;35:4027-34.
6. Lu S, Wu L, Jian H, et al. Sintilimab plus chemotherapy for patients with EGFR-mutated non-squamous non-small-cell lung cancer with disease progression after EGFR tyrosine-kinase inhibitor therapy (ORIENT-31): second interim analysis from a double-blind, randomised, placebo-controlled, phase 3 trial. Lancet Respir Med 2023;11:624-36.
7. Zhong R-B, Xu J-L, Lou Y-Q, Chu T-Q, Zhong H, Han B-H. Anlotinib or platinum-pemetrexed as second-line therapy in EGFR T790M-negative lung cancer. Ann Palliat Med 2020;9:1681-7.
8. HARMONi-A Study Investigators. Ivonescimab plus chemotherapy in non-small cell lung cancer with EGFR variant: a randomized clinical trial. JAMA 2024;332:561-70.
9. Passaro A, Wang J, Wang Y, et al. Amivantamab plus chemotherapy with and without lazertinib in EGFR-mutant advanced NSCLC after disease progression on osimertinib: primary results from the phase III MARIPOSA-2 study. Ann Oncol 2024;35:77-90.
10. Mok TS, Yu HA, Lim SM, et al. Patritumab deruxtecan (HER3-DXd) in resistant *EGFR*-mutated (*EGFR* m) advanced non-small cell lung cancer (NSCLC) after a third-generation EGFR TKI: the phase 3 HERTHENA-Lung02 study. J Clin Oncol 2025;43:Suppl 16:8506. abstract (https://ascopubs.org/doi/10.1200/JCO.2025.43.16_suppl.8506)
11. Sun X, Jia L, Wang T, et al. Trop2 binding IGF2R induces gefitinib resistance in NSCLC by remodeling the tumor microenvironment. J Cancer 2021;12:5310-9.
12. Baldacci S, Brea EJ, Facchinetti F, et al. Eradicating drug tolerant persister cells in EGFR-mutated non-small cell lung cancer by targeting TROP2 with CAR-T cellular therapy. Cancer Discov 2025 August 6 (Epub ahead of print).
13. Zhao S, Cheng Y, Wang Q, et al. Sacituzumab tirumotecan in advanced non-small-cell lung cancer with or without EGFR mutations: phase 1/2 and phase 2 trials. Nat Med 2025;31:1976-86.
14. Fang W, Li X, Wang Q, et al. Sacituzumab tirumotecan versus docetaxel for previously treated *EGFR*-mutated advanced non-small cell lung cancer: multicentre, open label, randomised controlled trial. BMJ 2025;389:e085680.
15. Aaronson NK, Ahmedzai S, Bergman B, et al. The European Organization for Research and Treatment of Cancer QLQ-C30: a quality-of-life instrument for use in international clinical trials in oncology. J Natl Cancer Inst 1993;85:365-76.
16. Fayers P, Bottomley A. Quality of life research within the EORTC — the EORTC QLQ-C30. Eur J Cancer 2002;38:Suppl 4:S125-S133.
17. Bergman B, Aaronson NK, Ahmedzai S, Kaasa S, Sullivan M. The EORTC QLQ-LC13: a modular supplement to the EORTC Core Quality of Life Questionnaire (QLQ-C30) for use in lung cancer clinical trials. Eur J Cancer 1994;30A:635-42.
18. Mok TS, Wu Y-L, Ahn M-J, et al. Osimertinib or platinum-pemetrexed in EGFR T790M-positive lung cancer. N Engl J Med 2017;376:629-40.
19. Paz-Ares LG, Juan-Vidal O, Mountzios GS, et al. Sacituzumab govitecan versus docetaxel for previously treated advanced or metastatic non-small cell lung cancer: the randomized, open-label phase III EVOKE-01 study. J Clin Oncol 2024;42:2860-72.
20. Ahn M-J, Tanaka K, Paz-Ares L, et al. Datopotamab deruxtecan versus docetaxel for previously treated advanced or metastatic non-small cell lung cancer: the randomized, open-label phase III TROPION-Lung01 study. J Clin Oncol 2025;43:260-72.
21. Sperduto PW, Yang TJ, Beal K, et al. Estimating survival in patients with lung cancer and brain metastases: an update of the Graded Prognostic Assessment for lung cancer using molecular markers (Lung-molGPA). JAMA Oncol 2017;3:827-31.
22. Soria J-C, Ohe Y, Vansteenkiste J, et al. Osimertinib in untreated *EGFR*-mutated advanced non–small-cell lung cancer. N Engl J Med 2018;378:113-25.
23. Lu S, Dong X, Jian H, et al. AENEAS: a randomized phase III trial of aumolertinib versus gefitinib as first-line therapy for locally advanced or metastaticnon-small-cell lung cancer with *EGFR* exon 19 deletion or L858R mutations. J Clin Oncol 2022;40:3162-71.
24. Shi Y, Chen G, Wang X, et al. Furmonertinib (AST2818) versus gefitinib as first-line therapy for Chinese patients with locally advanced or metastatic EGFR mutation-positive non-small-cell lung cancer (FURLONG): a multicentre, double-blind, randomised phase 3 study. Lancet Respir Med 2022;10:1019-28.

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25. Yin Y, Fan Y, Ouyang Q, et al. Sacituzumab tirumotecan in previously treated metastatic triple-negative breast cancer: a randomized phase 3 trial. Nat Med 2025;31:1969-75.
26. Ouyang Q, Rodon J, Liang Y, et al. Results of a phase 1/2 study of sacituzumab tirumotecan in patients with unresectable locally advanced or metastatic solid tumors refractory to standard therapies. J Hematol Oncol 2025;18:61.
27. Wu Y-L, Xiao Z, Li J, et al. Association of TROP2 quantitative continuous scoring (QCS) normalised membrane ratio (NMR) with efficacy in Chinese patients (pts) with advanced/metastatic non-small cell lung cancer (a/mNSCLC) treated with datopotamab deruxtecan (Dato-DXd) in TROPION-PanTumor02. J Thorac Oncol 2025;20:Suppl 1:S212-S213. abstract (https://www.jto.org/article/S1556-0864(25)00549-0/fulltext).

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